Gotta Catch 'Em All: Comprehensive Protein Annotation in the Parkinson's Microbiome Using BATH and Prokka

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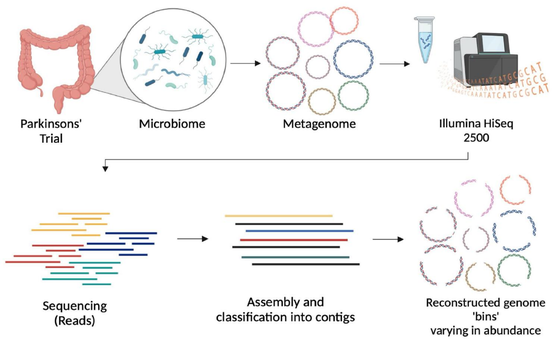
ABSTRACT

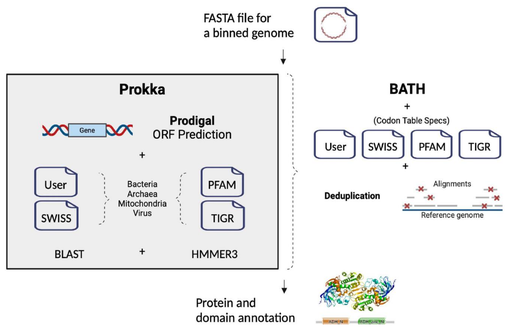
Genome annotation involves identifying and labeling various genomic features, a critical step in understanding the functional and regulatory mechanisms encoded within DNA. In annotation of microbial protein-coding DNA, significant challenges are posed by high sequence diversity, complexities of translation initiation mechanisms, and frameshifts (caused by mutation or sequencing error) that alter open reading frames (ORFs). These challenges become even more pronounced when analyzing complex microbiomes, where DNA from multiple organisms is entangled, making the environment particularly susceptible to assembly errors and sequencing error due to low read coverage.

In this study, we annotate a Parkinson's microbiome genomic sequence data set by aligning to a comprehensive database of protein families using a combination of Prokka and our software for annotating protein-coding DNA, called BATH (poster A-041). Prokka annotates protein-coding DNA by detecting ORFs using Prodigal, then comparing the resulting sequences using a combination of BLAST+blastp and HMMER3 (depending on underlying sequence library). BATH is a new tool designed to improve annotation of protein-coding DNA through accurate detection of frameshifts.

To identify strengths and limitations of the two tools, we performed a detailed comparison of their annotation results. Thanks to its frameshift-aware sequence alignment, BATH annotates a substantial number of ORFs that Prokka does not. Additionally, BATH also detects new proteins that Prodigal misses. This is likely due to Prokka's limitation in accommodating frameshifts or variations in start sites, which BATH can accommodate more flexibly. Our study highlights BATH as a robust solution for the challenges posed by genome annotation for microbial communities due to its frameshift-aware sensitive pHMM annotation mechanisms.

METHODOLOGY

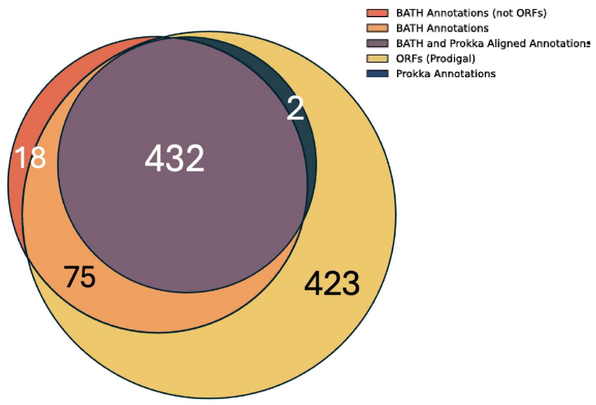




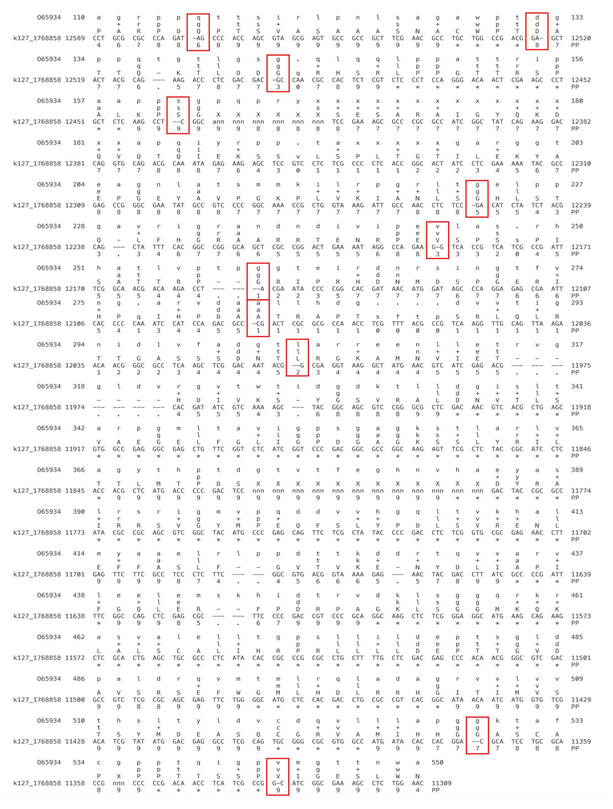
RESULTS

1. BATH uncovers additional proteins beyond Prokka and Prodigal ORFs

1.A. Overlaps between BATH, Prokka, Prodigal annotation

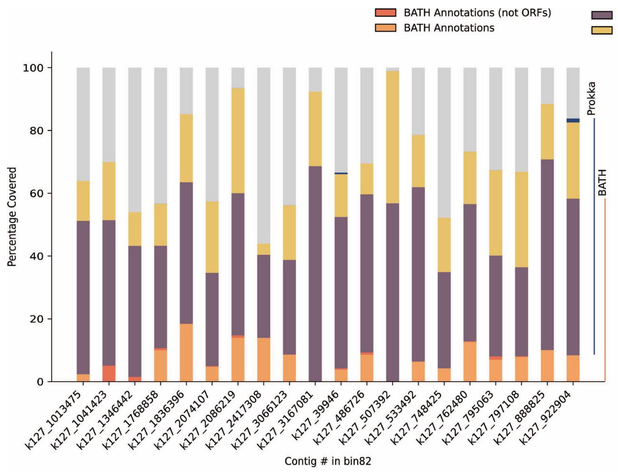


1.C. An example of BATH alignment with frameshifts

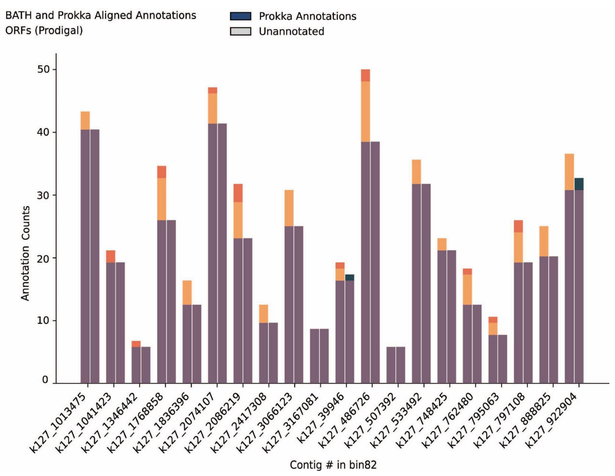


2. BATH improves annotation count and coverage per contig.

2.A. BATH improves annotation coverage for bin82 contigs

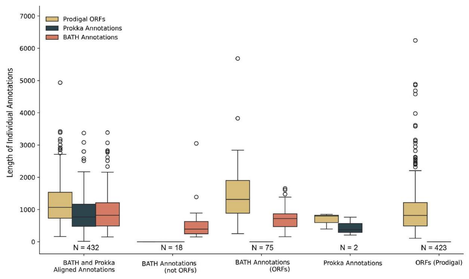


2.B. BATH annotates additional proteins for each contig of bin82

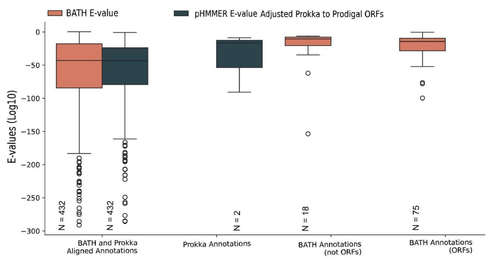


3. BATH alignments closely align with Prokka and Prodigal ORFs

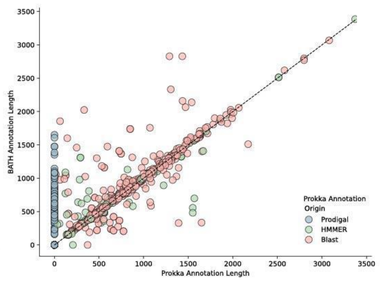
3.A. Length distributions



3.B. E-values

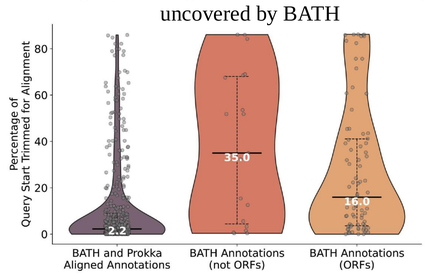


3.C. Annotations lengths

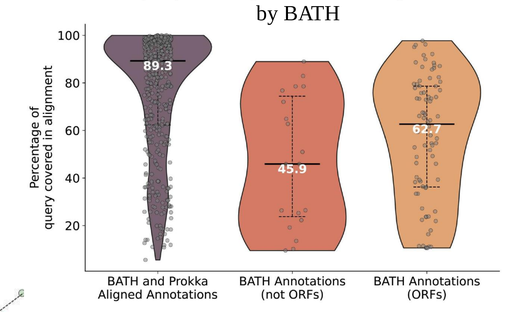


4. BATH uncovers new annotations by bypassing traditional ORF detection.

4.A. Larger percentage of start is trimmed in alignments uncovered by BATH



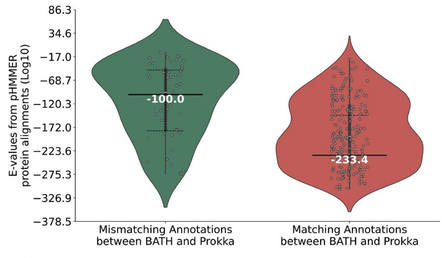
4.B. Lower query coverage is found in alignments uncovered by BATH



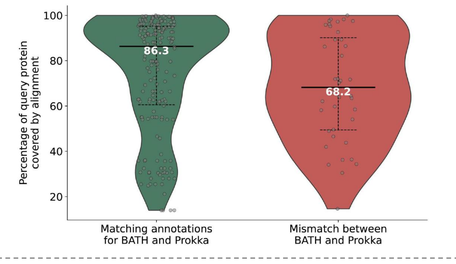
5. BATH and Prokka produce the same protein / COG level annotations for matching alignments

5.A. Percentage of matching annotations

5.B. Mismatching annotations show high sequence similarity



5.C. Mismatching annotations show shorter query coverage



6. Conclusion and Future Directions

We demonstrate that BATH yields effective protein annotations for microbiome data by:

- Providing larger contig coverage
- Annotating additional proteins
- Achieving greater ORF coverage, indicating enhanced alignment accuracy
- Detecting annotated proteins without ORF prediction, highlighting increased sensitivity
- Producing annotations with comparable E-values and lengths to Prokka

7. References

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4. Genevieve R Krause, Walt Shands, Travis J Wheeler, **Sensitive and error-tolerant annotation of protein-coding DNA with BATH**, Bioinformatics Advances, 2024; vbae088, https://doi.org/10.1093/bioadv/vbae088

8. Acknowledgements

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